YOLO26s 1,400-Epoch Controlled Ablation Study

Full methods, per-variant changes, validation statistics, and class-wise results

Prepared 4 November 2026 | 14 runs x 100 epochs = 1,400 completed training epochs

Main finding. Among the 13 tested changes, input upscaling from 640 to 800 pixels was the only variant that improved all five reported metrics against the baseline on this validation split: precision +0.36 pp, recall +0.83 pp, derived F1 +0.68 pp, mAP50 +0.57 pp, and mAP50-95 +0.18 pp. These are modest single-seed validation gains; they are a candidate for confirmation, not proof of general improvement.

Scope and interpretation

This report covers the completed YOLO26s ablation queue dated 21 October 2026. Each arm trained for 100 epochs with seed 42, for 1,400 training epochs total. Metrics are from the fixed validation split and the best checkpoint selected by validation mAP50-95, then re-evaluated on validation. The held-out internal test split was not used for model or hyperparameter selection. Follow-on multi-seed repeats are a separate experiment and are not included here.

The five headline values are precision, recall, F1, mAP50, and mAP50-95. F1 is calculated from the aggregate precision and recall values and is labeled as derived; it is not a separately recorded evaluator output. Differences are percentage points (pp), not percent relative change.

1. Dataset and baseline procedure

Dataset: N-RDD2024 version 3, prepared as a ten-class detection dataset. The fixed split contains 13,782 training images, 3,704 validation images, and 1,509 internal test images. The study used training and validation only. The rare D90 (rutting) class has 40 training boxes, 3 validation boxes, and 1 internal-test box, so its class-wise scores are highly uncertain.

Baseline training recipe

Setting	Baseline value	Purpose / control
Model start	Pretrained YOLO26s (yolo26s.pt), transferred to 10 classes	Same starting detector for every arm; architecture variants transfer compatible weights.
Training	100 epochs; patience 0; seed 42; deterministic flag enabled	Fixed compute budget; no early stopping.
Image / batch	640 x 640; batch 4	Resolution arm changes only image size to 800.
Compute	RTX 4060; AMP on; workers 0; cache off	Same local GPU and data-loading choices.
Optimizer	MuSGD; lr0 0.010; momentum 0.9; weight decay 0.0005	Optimizer and schedule held fixed outside the named LR / WD arms.
Augmentation	Mosaic 1.0; close mosaic at epoch 10; standard Ultralytics augmentations	Mosaic arms change its probability or shutoff point.
Checkpoint selection	Highest validation mAP50-95; best.pt re-evaluated on validation	No test-set tuning or selection.

Model footprint and baseline checkpoint

Baseline model size: 9,955,604 parameters; 22.78 GFLOPs at 640. Baseline best epoch: 53; latest trained epoch: 100. Baseline checkpoint SHA-256: 74a4853cdb86dd5182a526d37710dd74daf32d5711714dc1a13f03e176306fb7.

Training procedure, step by step

1. Load the pretrained YOLO26s small model and adapt its detection layer to the ten N-RDD2024 classes.
2. Read the fixed training split; apply the same geometric and photometric training augmentations and mosaic policy specified above.
3. For each batch, run the detector forward pass, compute the standard YOLO detection losses, and update model weights with MuSGD under automatic mixed precision.
4. At each epoch, evaluate on the fixed validation split and save the training history and checkpoints.
5. After 100 epochs, select best.pt using validation mAP50-95 and re-run validation to record aggregate and class-wise precision, recall, AP50, and AP50-95.
6. Compare each one-factor change with the control. The internal test split is kept out of this comparison.

2. Each ablation change and outcome

The table reports the best-validation checkpoint metrics for every 100-epoch run. The baseline is the reference row. F1 is derived from aggregate P and R. All deltas are against control and shown in percentage points.

Run / intervention	Specific change from control	P	R	F1*	mAP50	mAP50-95	Delta: P / R / F1 / AP50 / AP95
Baseline control best epoch 53	Reference recipe; no intervention.	68.82	49.66	57.69	54.02	31.17	+0.00 / +0.00 / +0.00 / +0.00 / +0.00
Learning rate 0.005 best epoch 62	Only lr0 changed: 0.010 to 0.005 (half the baseline initial learning rate).	69.83	49.01	57.59	53.19	30.50	+1.02 / -0.66 / -0.10 / -0.83 / -0.67
Learning rate 0.020 best epoch 65	Only lr0 changed: 0.010 to 0.020 (double the baseline initial learning rate).	68.53	49.67	57.60	53.98	30.83	-0.29 / +0.01 / -0.09 / -0.04 / -0.34
Weight decay 0.0001 best epoch 53	Only weight_decay changed: 0.0005 to 0.0001.	68.61	50.31	58.05	53.62	30.77	-0.21 / +0.64 / +0.36 / -0.40 / -0.40
Weight decay 0.0010 best epoch 63	Only weight_decay changed: 0.0005 to 0.0010.	69.96	48.40	57.22	53.47	30.77	+1.14 / -1.26 / -0.47 / -0.55 / -0.40
Mosaic probability 0.25 best epoch 48	Only mosaic changed: probability 1.00 to 0.25.	67.29	48.57	56.42	51.83	29.30	-1.53 / -1.09 / -1.27 / -2.19 / -1.87
Mosaic probability 0.50 best epoch 63	Only mosaic changed: probability 1.00 to 0.50.	68.89	49.11	57.34	52.95	29.93	+0.07 / -0.55 / -0.35 / -1.07 / -1.24
Close mosaic at 20 epochs best epoch 53	Only close_mosaic changed: mosaic is disabled for the final 20 epochs instead of 10.	68.82	49.66	57.69	54.02	31.17	+0.00 / +0.00 / +0.00 / +0.00 / +0.00
Input resolution 800 best epoch 61	Only imgsz changed: 640 to 800 pixels; batch size and all other configured settings held fixed.	69.18	50.49	58.37	54.59	31.36	+0.36 / +0.83 / +0.68 / +0.57 / +0.18
Noise/blur probability 0.10 best epoch 66	Added an on-device per-image corruption with probability 0.10. Selected images received Gaussian noise (sigma sampled 0.005-0.03) or a 3x3 Gaussian blur, each with equal chance; all other settings were held fixed.	69.55	50.05	58.21	53.74	30.94	+0.73 / +0.39 / +0.52 / -0.28 / -0.23
Noise/blur probability 0.25 best epoch 67	Same mixed Gaussian-noise/3x3-blur training augmentation as noise_010, with probability 0.25 instead of 0.10.	69.65	50.05	58.25	53.35	30.60	+0.83 / +0.39 / +0.56 / -0.67 / -0.57
Added P2 detection branch best epoch 64	Replaced the standard detection head with a four-scale head adding a stride-4 P2 prediction branch; pretrained tensors were transferred where shapes matched and new/mismatched tensors initialized by the model. The selected recipe was the baseline control, so optimizer and augmentation settings remained at baseline values.	68.84	49.12	57.33	53.77	30.92	+0.03 / -0.54 / -0.36 / -0.25 / -0.25
Learned weighted fusion best epoch 53	Replaced ordinary feature concatenation nodes with trainable normalized positive branch weights before concatenation. This is weighted concatenation, not a full BiFPN. The selected recipe was the baseline control, so optimizer and augmentation settings remained at baseline values.	69.73	49.04	57.58	53.77	30.70	+0.92 / -0.62 / -0.11 / -0.25 / -0.47
P2 branch + learned fusion best epoch 52	Combined the added stride-4 P2 detection branch with the trainable weighted-concatenation nodes. The selected recipe was the baseline control, so optimizer and augmentation settings remained at baseline values.	67.09	49.31	56.84	53.32	30.74	-1.73 / -0.35 / -0.85 / -0.70 / -0.43

* F1 derived from aggregate precision and recall. AP95 denotes mAP50-95.

How to read the comparisons

Resolution 800 improved all five headline measures against control in this seed-42 validation run. Noise/blur augmentation increased precision, recall, and derived F1 but lowered both mAP measures. The remaining interventions traded off measures or reduced AP; none improved all five. The close-mosaic-20 run selected epoch 53, before either run's mosaic shutoff point, which explains why its selected checkpoint is effectively the same training regime as control up to selection.

3. Full training and validation statistics

This section adds best epoch, final epoch, training duration recorded in results.csv, model footprint, inference timing, and checkpoint identity for every run.

Run	Epochs	Best ep.	Train h*	Params	GFLOPs	Pre ms	Infer ms	Post ms	Checkpoint SHA-256
Baseline control	100	53	14.4	9,955,604	22.78	0.27	5.00	0.80	74a4853cdb86dd5182a526d37710dd74daf32d5711714dc1a13f03e176306fb7
Learning rate 0.005	100	62	12.7	9,955,604	22.78	0.26	5.04	0.80	0914dcf841aa463ff510d5b5645e64d489e43216d713c8dd356c267a0be50deb
Learning rate 0.020	100	65	14.3	9,955,604	22.78	0.28	4.97	0.94	3c7904d9e919202c4bddb6c2f2adb954fc660fc85d8d4f413eee480a5156f0e3
Weight decay 0.0001	100	53	18.8	9,955,604	22.78	0.28	4.97	0.95	e1c9c31280169d87c53d16ada5c225013cd230de84a2e2b05ecbf378a2fb4f90
Weight decay 0.0010	100	63	16.5	9,955,604	22.78	0.29	5.01	0.97	8f57f440283425a3d3d224fe843fc6527452219b334ee27dc917df0fe8a024ec
Mosaic probability 0.25	100	48	22.1	9,955,604	22.78	0.31	6.20	1.42	0caadcc7db0efd89eb430512b5d69e8d7087b13c5bbee1aa295265f2f89d66a5
Mosaic probability 0.50	100	63	22.7	9,955,604	22.78	0.29	4.97	1.20	65de61550dfee1c7e56232b2dcd11697f9afd483da758ca80a2975f863d2e582
Close mosaic at 20 epochs	100	53	19.4	9,955,604	22.78	0.30	4.98	1.20	eb3e18bdded1fc993ae66f04fff734dfc5f3c5df3dcee8c2eddb65f17c0227b8
Input resolution 800	100	61	24.2	9,955,604	35.81	0.43	7.74	1.19	fe746fc547344e50f6344f3cc588c1c9bc06653ec2c4de203eb23a8ad64c7013
Noise/blur probability 0.10	100	66	21.8	9,955,604	22.78	0.33	4.99	1.23	4bdc29209211111138de76781180c359cdd67e8ff5d61c8788bc8aee134db243c
Noise/blur probability 0.25	100	67	11.1	9,955,604	22.78	0.26	5.34	0.78	eb65c02cbf0866966101b7b200126280cc56f17df72598bd4d95d52a5007d77d
Added P2 detection branch	100	64	16.8	9,535,920	24.98	0.27	6.88	0.94	a9919f8f01ce8ca3a5a9ac3b89a43f71c8af2d5115bd8b2628b0c0fd398c86cb
Learned weighted fusion	100	53	17.1	9,955,612	22.78	0.28	5.18	0.97	81d6e44874d13753120f6d5b2fc791bfa60d7a5376cd500b6bac377ee816121d
P2 branch + learned fusion	100	52	18.5	9,535,930	24.98	0.27	7.40	1.03	e5ef83af4a923fc654d3b2db012d50669dba656843002e35f099d5dd776e43b8

* Training hours are taken from the recorded epoch-100 results.csv time field; they exclude any subsequent standalone validation timing. Per-image speed is from the separate validation pass and is hardware/runtime dependent. Model parameter/GFLOP values are the stored run-time measurements where available. SHA-256 identifies each selected best checkpoint.

Final-epoch versus selected-checkpoint validation

Run	Selected ep.	Selected mAP50	Selected mAP50-95	Epoch 100 mAP50	Epoch 100 mAP50-95
Baseline control	53	54.02	31.17	52.90	30.20
Learning rate 0.005	62	53.19	30.50	51.93	29.62
Learning rate 0.020	65	53.98	30.83	53.72	30.55
Weight decay 0.0001	53	53.62	30.77	52.21	29.79
Weight decay 0.0010	63	53.47	30.77	52.31	29.86
Mosaic probability 0.25	48	51.83	29.30	50.20	28.09

Run	Selected ep.	Selected mAP50	Selected mAP50-95	Epoch 100 mAP50	Epoch 100 mAP50-95
Mosaic probability 0.50	63	52.95	29.93	51.29	29.11
Close mosaic at 20 epochs	53	54.02	31.17	52.62	30.06
Input resolution 800	61	54.59	31.36	52.86	30.30
Noise/blur probability 0.10	66	53.74	30.94	52.67	30.33
Noise/blur probability 0.25	67	53.35	30.60	52.21	29.89
Added P2 detection branch	64	53.77	30.92	52.91	30.11
Learned weighted fusion	53	53.77	30.70	53.35	30.14
P2 branch + learned fusion	52	53.32	30.74	51.85	29.52

4. Class-wise validation statistics

All recorded best-checkpoint class metrics are listed below. Supports are counts of ground-truth boxes in the fixed validation label files. P, R, AP50, and AP50-95 are class-specific. Values are percentages. The D90 result has only three validation boxes and must not be interpreted as a stable estimate.

Run	Class	Defect	GT boxes	P %	R %	AP50 %	AP50-95 %
Baseline control	D00	Longitudinal crack	4632	67.03	52.81	59.24	33.01
Baseline control	D10	Transverse crack	1991	61.87	54.65	58.04	28.70
Baseline control	D20	Alligator crack	1466	68.37	58.24	63.63	34.34
Baseline control	D30	Repaired crack	106	64.12	35.41	41.16	23.24
Baseline control	D40	Pothole	728	63.34	49.18	52.33	24.32
Baseline control	D50	Pedestrian crossing blur	159	62.93	63.52	65.57	42.03
Baseline control	D60	Lane line blur	1027	55.54	49.17	50.04	28.22
Baseline control	D70	Manhole cover	698	79.76	76.79	83.39	47.06
Baseline control	D80	Patchy road section	201	65.20	56.85	66.83	50.78
Baseline control	D90	Rutting	3	100.00	0.00	0.00	0.00
Learning rate 0.005	D00	Longitudinal crack	4632	71.02	52.18	59.54	33.27
Learning rate 0.005	D10	Transverse crack	1991	66.47	50.98	57.78	28.76
Learning rate 0.005	D20	Alligator crack	1466	70.00	58.73	63.51	34.13
Learning rate 0.005	D30	Repaired crack	106	48.71	36.79	35.78	20.67
Learning rate 0.005	D40	Pothole	728	67.54	47.94	52.82	23.49
Learning rate 0.005	D50	Pedestrian crossing blur	159	69.18	62.89	66.99	42.37
Learning rate 0.005	D60	Lane line blur	1027	58.02	47.91	49.40	27.71
Learning rate 0.005	D70	Manhole cover	698	82.90	76.41	82.82	46.46
Learning rate 0.005	D80	Patchy road section	201	64.47	56.22	63.17	48.08
Learning rate 0.005	D90	Rutting	3	100.00	0.00	0.07	0.04
Learning rate 0.020	D00	Longitudinal crack	4632	68.64	52.25	59.59	33.39
Learning rate 0.020	D10	Transverse crack	1991	63.83	53.44	58.35	28.63
Learning rate 0.020	D20	Alligator crack	1466	69.47	58.37	64.62	35.21
Learning rate 0.020	D30	Repaired crack	106	56.05	40.91	42.97	21.67
Learning rate 0.020	D40	Pothole	728	66.95	46.47	52.62	24.38
Learning rate 0.020	D50	Pedestrian crossing blur	159	61.08	60.38	65.52	40.98
Learning rate 0.020	D60	Lane line blur	1027	58.07	48.30	49.67	28.11
Learning rate 0.020	D70	Manhole cover	698	80.74	76.93	82.11	46.25
Learning rate 0.020	D80	Patchy road section	201	60.46	59.70	64.40	49.71
Learning rate 0.020	D90	Rutting	3	100.00	0.00	0.00	0.00
Weight decay 0.0001	D00	Longitudinal crack	4632	67.33	53.48	59.30	33.10
Weight decay 0.0001	D10	Transverse crack	1991	64.35	54.50	58.84	28.90
Weight decay 0.0001	D20	Alligator crack	1466	68.51	58.19	63.81	34.27
Weight decay 0.0001	D30	Repaired crack	106	49.39	37.74	35.91	19.44
Weight decay 0.0001	D40	Pothole	728	69.63	46.15	54.21	24.72
Weight decay 0.0001	D50	Pedestrian crossing blur	159	65.16	65.88	66.22	42.46
Weight decay 0.0001	D60	Lane line blur	1027	57.59	49.72	50.09	28.63

Run	Class	Defect	GT boxes	P %	R %	AP50 %	AP50-95 %
Weight decay 0.0001	D70	Manhole cover	698	80.36	76.20	81.68	46.54
Weight decay 0.0001	D80	Patchy road section	201	63.74	61.22	66.15	49.62
Weight decay 0.0001	D90	Rutting	3	100.00	0.00	0.00	0.00
Weight decay 0.0010	D00	Longitudinal crack	4632	70.18	51.99	59.49	33.37
Weight decay 0.0010	D10	Transverse crack	1991	64.87	51.68	58.07	28.70
Weight decay 0.0010	D20	Alligator crack	1466	71.80	58.69	64.93	35.06
Weight decay 0.0010	D30	Repaired crack	106	59.47	33.23	37.36	20.45
Weight decay 0.0010	D40	Pothole	728	66.93	46.15	52.06	23.55
Weight decay 0.0010	D50	Pedestrian crossing blur	159	60.79	59.75	64.29	41.13
Weight decay 0.0010	D60	Lane line blur	1027	58.60	49.27	50.21	28.08
Weight decay 0.0010	D70	Manhole cover	698	81.32	76.36	82.61	46.65
Weight decay 0.0010	D80	Patchy road section	201	65.59	56.91	65.65	50.69
Weight decay 0.0010	D90	Rutting	3	100.00	0.00	0.00	0.00
Mosaic probability 0.25	D00	Longitudinal crack	4632	67.65	52.68	58.79	32.79
Mosaic probability 0.25	D10	Transverse crack	1991	63.93	51.78	57.57	28.41
Mosaic probability 0.25	D20	Alligator crack	1466	69.75	57.98	62.88	33.30
Mosaic probability 0.25	D30	Repaired crack	106	43.27	33.02	31.85	16.92
Mosaic probability 0.25	D40	Pothole	728	65.05	47.39	52.33	22.88
Mosaic probability 0.25	D50	Pedestrian crossing blur	159	59.12	62.26	63.41	40.09
Mosaic probability 0.25	D60	Lane line blur	1027	58.05	49.46	48.83	27.35
Mosaic probability 0.25	D70	Manhole cover	698	79.45	76.44	81.82	44.80
Mosaic probability 0.25	D80	Patchy road section	201	66.62	54.73	60.86	46.47
Mosaic probability 0.25	D90	Rutting	3	100.00	0.00	0.00	0.00
Mosaic probability 0.50	D00	Longitudinal crack	4632	68.87	52.33	59.30	33.32
Mosaic probability 0.50	D10	Transverse crack	1991	64.79	53.05	58.18	28.60
Mosaic probability 0.50	D20	Alligator crack	1466	68.27	57.23	63.17	33.95
Mosaic probability 0.50	D30	Repaired crack	106	54.98	37.74	37.25	17.93
Mosaic probability 0.50	D40	Pothole	728	66.50	47.66	51.45	23.06
Mosaic probability 0.50	D50	Pedestrian crossing blur	159	61.42	61.01	65.48	41.91
Mosaic probability 0.50	D60	Lane line blur	1027	58.50	47.91	49.15	27.42
Mosaic probability 0.50	D70	Manhole cover	698	81.59	74.94	82.88	45.79
Mosaic probability 0.50	D80	Patchy road section	201	63.98	59.20	62.36	47.34
Mosaic probability 0.50	D90	Rutting	3	100.00	0.00	0.26	0.03
Close mosaic at 20 epochs	D00	Longitudinal crack	4632	67.03	52.81	59.24	33.01
Close mosaic at 20 epochs	D10	Transverse crack	1991	61.87	54.65	58.04	28.70
Close mosaic at 20 epochs	D20	Alligator crack	1466	68.37	58.24	63.63	34.34
Close mosaic at 20 epochs	D30	Repaired crack	106	64.12	35.41	41.16	23.24
Close mosaic at 20 epochs	D40	Pothole	728	63.34	49.18	52.33	24.32
Close mosaic at 20 epochs	D50	Pedestrian crossing blur	159	62.93	63.52	65.57	42.03
Close mosaic at 20 epochs	D60	Lane line blur	1027	55.54	49.17	50.04	28.22
Close mosaic at 20 epochs	D70	Manhole cover	698	79.76	76.79	83.39	47.06
Close mosaic at 20 epochs	D80	Patchy road section	201	65.20	56.85	66.83	50.78

Run	Class	Defect	GT boxes	P %	R %	AP50 %	AP50-95 %
Close mosaic at 20 epochs	D90	Rutting	3	100.00	0.00	0.00	0.00
Input resolution 800	D00	Longitudinal crack	4632	70.12	52.63	60.20	33.58
Input resolution 800	D10	Transverse crack	1991	65.89	52.68	58.63	29.07
Input resolution 800	D20	Alligator crack	1466	69.00	60.10	65.06	34.75
Input resolution 800	D30	Repaired crack	106	54.99	40.57	43.97	24.36
Input resolution 800	D40	Pothole	728	65.28	48.76	53.96	25.12
Input resolution 800	D50	Pedestrian crossing blur	159	63.29	63.97	67.23	43.04
Input resolution 800	D60	Lane line blur	1027	58.93	50.16	50.92	28.34
Input resolution 800	D70	Manhole cover	698	81.12	77.79	82.97	47.24
Input resolution 800	D80	Patchy road section	201	63.15	58.21	62.96	48.06
Input resolution 800	D90	Rutting	3	100.00	0.00	0.00	0.00
Noise/blur probability 0.10	D00	Longitudinal crack	4632	69.35	52.37	59.94	33.74
Noise/blur probability 0.10	D10	Transverse crack	1991	62.67	53.04	57.92	28.94
Noise/blur probability 0.10	D20	Alligator crack	1466	68.05	59.28	64.67	35.03
Noise/blur probability 0.10	D30	Repaired crack	106	53.19	39.62	34.18	18.65
Noise/blur probability 0.10	D40	Pothole	728	66.33	47.12	52.64	24.63
Noise/blur probability 0.10	D50	Pedestrian crossing blur	159	68.60	64.15	69.82	43.39
Noise/blur probability 0.10	D60	Lane line blur	1027	57.46	51.31	50.09	28.43
Noise/blur probability 0.10	D70	Manhole cover	698	81.69	76.22	82.97	46.83
Noise/blur probability 0.10	D80	Patchy road section	201	68.13	57.44	65.08	49.74
Noise/blur probability 0.10	D90	Rutting	3	100.00	0.00	0.09	0.05
Noise/blur probability 0.25	D00	Longitudinal crack	4632	69.14	52.55	59.63	33.30
Noise/blur probability 0.25	D10	Transverse crack	1991	63.66	54.65	58.62	29.41
Noise/blur probability 0.25	D20	Alligator crack	1466	68.32	60.10	64.63	35.23
Noise/blur probability 0.25	D30	Repaired crack	106	57.89	36.31	36.15	20.33
Noise/blur probability 0.25	D40	Pothole	728	66.43	47.12	51.88	23.57
Noise/blur probability 0.25	D50	Pedestrian crossing blur	159	65.66	66.04	67.74	42.25
Noise/blur probability 0.25	D60	Lane line blur	1027	57.95	50.59	49.59	27.56
Noise/blur probability 0.25	D70	Manhole cover	698	82.07	74.10	82.31	46.42
Noise/blur probability 0.25	D80	Patchy road section	201	65.34	59.09	62.69	47.88
Noise/blur probability 0.25	D90	Rutting	3	100.00	0.00	0.29	0.03
Added P2 detection branch	D00	Longitudinal crack	4632	68.39	52.16	59.41	33.40
Added P2 detection branch	D10	Transverse crack	1991	62.57	53.34	57.82	28.28
Added P2 detection branch	D20	Alligator crack	1466	67.87	59.35	64.12	34.41
Added P2 detection branch	D30	Repaired crack	106	57.46	41.51	42.67	23.68
Added P2 detection branch	D40	Pothole	728	66.91	46.29	52.36	23.81
Added P2 detection branch	D50	Pedestrian crossing blur	159	62.14	59.75	63.95	41.31
Added P2 detection branch	D60	Lane line blur	1027	57.72	49.05	50.27	27.84
Added P2 detection branch	D70	Manhole cover	698	81.93	74.07	81.71	46.52
Added P2 detection branch	D80	Patchy road section	201	63.43	55.72	65.36	49.99
Added P2 detection branch	D90	Rutting	3	100.00	0.00	0.00	0.00
Learned weighted fusion	D00	Longitudinal crack	4632	69.19	51.45	59.15	33.05

Run	Class	Defect	GT boxes	P %	R %	AP50 %	AP50-95 %
Learned weighted fusion	D10	Transverse crack	1991	65.83	52.64	58.95	28.94
Learned weighted fusion	D20	Alligator crack	1466	68.61	58.25	63.74	34.22
Learned weighted fusion	D30	Repaired crack	106	52.00	33.96	34.39	18.08
Learned weighted fusion	D40	Pothole	728	67.17	43.96	51.49	23.80
Learned weighted fusion	D50	Pedestrian crossing blur	159	64.97	64.78	66.93	42.44
Learned weighted fusion	D60	Lane line blur	1027	59.85	50.83	50.79	28.24
Learned weighted fusion	D70	Manhole cover	698	80.52	75.82	83.74	46.61
Learned weighted fusion	D80	Patchy road section	201	69.19	58.71	66.59	51.30
Learned weighted fusion	D90	Rutting	3	100.00	0.00	1.97	0.32
P2 branch + learned fusion	D00	Longitudinal crack	4632	67.72	52.31	58.97	32.84
P2 branch + learned fusion	D10	Transverse crack	1991	60.80	52.49	56.93	27.89
P2 branch + learned fusion	D20	Alligator crack	1466	68.77	58.88	65.07	34.38
P2 branch + learned fusion	D30	Repaired crack	106	44.87	35.85	36.60	20.58
P2 branch + learned fusion	D40	Pothole	728	67.32	47.25	53.10	25.25
P2 branch + learned fusion	D50	Pedestrian crossing blur	159	62.38	61.64	66.16	41.77
P2 branch + learned fusion	D60	Lane line blur	1027	57.23	49.37	49.54	28.33
P2 branch + learned fusion	D70	Manhole cover	698	79.25	75.50	82.33	47.27
P2 branch + learned fusion	D80	Patchy road section	201	62.54	59.80	64.50	49.08
P2 branch + learned fusion	D90	Rutting	3	100.00	0.00	0.00	0.00

5. Interpretation, limitations, and next step

What this experiment supports

At seed 42, the 800-pixel input variant is the only tested change whose best validation checkpoint exceeds the baseline in precision, recall, derived F1, mAP50, and mAP50-95 at the same time. The measured margins are small: +0.36 pp precision, +0.83 pp recall, +0.69 pp F1, +0.57 pp mAP50, and +0.19 pp mAP50-95. Higher input resolution is therefore the sole promising candidate from this study, subject to matched-seed confirmation.

What it does not establish

This is a single fixed split and one training seed. Small gains can be within run-to-run variation and do not demonstrate statistical significance. Validation results are not an independent test result. D90 has extremely low support. This experiment also does not establish a research novelty claim or superiority to published scores obtained with different label sets, splits, or evaluation protocols.

Recommended confirmation

Complete the already planned paired 640-versus-800 repeat runs on additional dispersed seeds, using the same fixed split, pretrained weights, training budget, and checkpoint-selection rule. Report paired per-seed deltas, the mean and standard deviation for all five metrics, compute cost, and inference speed. Keep the held-out test set untouched until the candidate and settings are locked; then evaluate the selected model and baseline once under the same protocol.

Reported metric definitions

Precision = TP/(TP+FP). Recall = TP/(TP+FN). F1 = 2PR/(P+R), calculated here from aggregate P and R. AP integrates precision-recall for one class at one IoU setting. mAP50 averages class AP at IoU 0.50; mAP50-95 averages AP over IoU 0.50 through 0.95 in 0.05 steps. Higher is better for the five detection metrics. Inference timing excludes camera capture, decoding, networking, and geolocation.

Experiment provenance

Study folder: experiments/yolo26s_ablation_20261021. Runner: run_yolo26_ablation.py with ablation_components.py for the added head/fusion modules. Dataset split-manifest SHA-256: 25a8ab1f25b485b3db90a6bcfd1d9ae6c44ccaa53479e8dc3561780f7b68817e. Baseline pretrained weights SHA-256: 646f8bc3fe0a656803d95c294f7852321748cb29d13466a1af8862e2db384a1b. Selected best-checkpoint SHA-256 values appear in Section 3. Results are validation-only and each run completed exactly 100 epochs.